

Agentic CI/CD

Student Lab Guide · GitHub Actions + LangChain + Gemini

Prerequisites

■ Make sure everything below is ready **before** the lab session starts.

Git Must be installed on your system. <https://git-scm.com/downloads>

```
$ git --version
```

GitHub Account Free account at github.com. No credit card needed. <https://github.com/signup>

Gemini API Key Free key from Google AI Studio. <https://aistudio.google.com>

Python 3.9+ Optional – only needed for local runs. <https://python.org/downloads>

```
$ python3 --version
```

Lab Steps

1 Fork the Repository

Open the link below in your browser and click **Fork** (top-right). This copies the project into your own GitHub account.

```
https://github.com/kumarsirish/agentic-cicd-github
```

✓ Keep the repository **public** so GitHub Actions can run on the free tier.

2 Clone Your Fork Locally

Open a terminal and run the commands below. Replace **YOUR-USERNAME** with your GitHub username.

```
$ git clone https://github.com/YOUR-USERNAME/agentic-cicd-github.git
$ cd agentic-cicd-github
```

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Add the Gemini API Secret

The pipeline reads GEMINI_API_KEY from a GitHub Environment named **my-secrets**. Follow the sub-steps below exactly.

- 3a

Go to your forked repo on GitHub.
- 3b

Click **Settings** (top menu of the repo).
- 3c

In the left sidebar click **Environments**.
- 3d

Click **New environment**, name it **my-secrets**, then click **Configure environment**.
- 3e

Scroll to **Environment secrets** → click **Add secret**.
- 3f

Name: GEMINI_API_KEY | Value: your key from AI Studio.
- 3g

Click **Add secret** to save.

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Trigger the Pipeline

Push any small change to the **main** branch – this automatically starts the GitHub Actions workflow.

```
# Example: update the README and push
$ echo "# My Run" >> README.md
$ git add README.md
$ git commit -m "trigger pipeline"
$ git push origin main
```

■ You can also trigger manually: **Actions** tab → **Agentic CI/CD** → **Run workflow** → **Run workflow**.

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Watch the Pipeline Run

Click the **Actions** tab in your repo. You will see the **Agentic CI/CD** workflow running.

Stage	What to look for
Install Dependencies	All packages installed without errors
Build	app.py compiles successfully
Test	pytest output captured in test-results.log
Upload Test Results	test-results artifact appears at bottom of run page
LangChain Agent Analysis	Agent Decision: GO / RETRY / STOP printed in logs
Decision Gate	■ GO continues ■ RETRY re-runs tests ■ STOP fails
Deploy	deploy.sh runs and prints Deployment completed

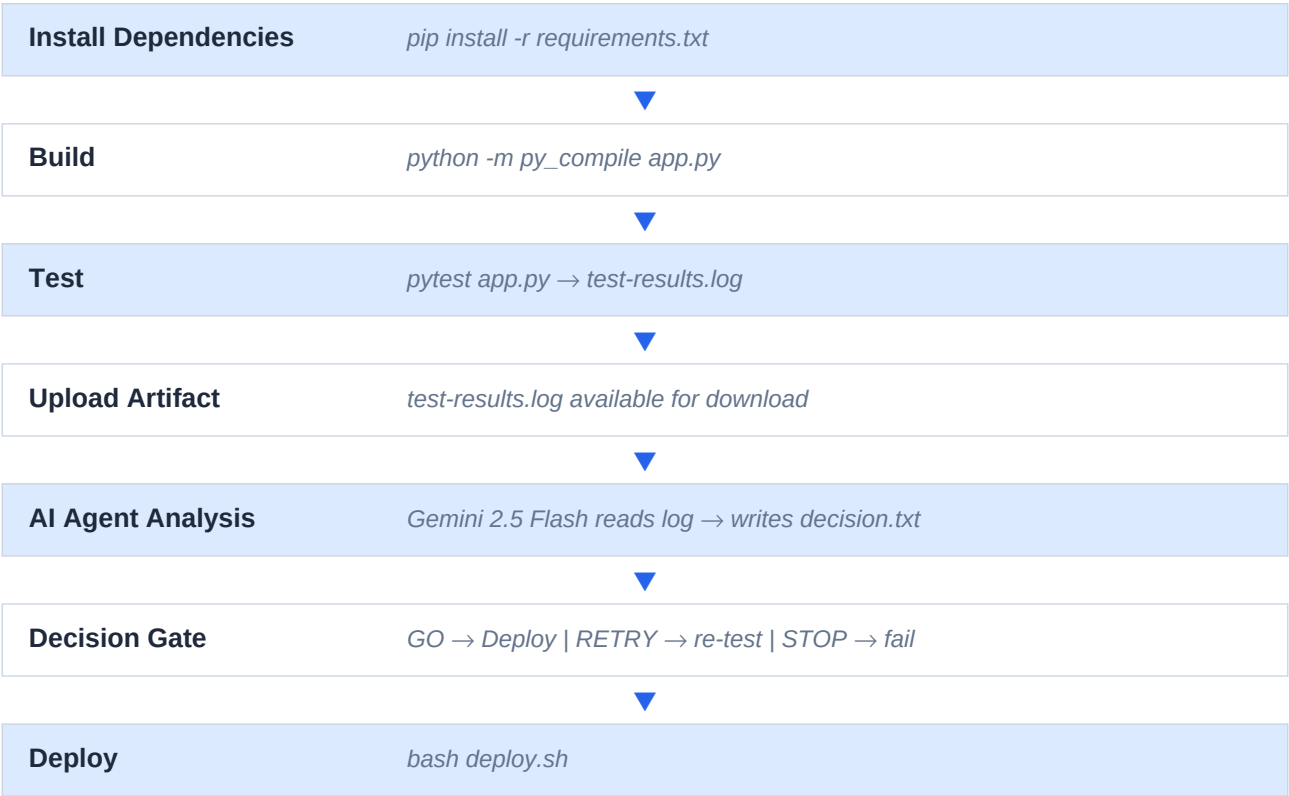
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Download the Test Results Artifact

At the bottom of the completed run page, find the **Artifacts** section and click **test-results** to download test-results.log.

■ The artifact is kept for **90 days** by default (GitHub free tier).

Pipeline Overview



Key Files in the Repository

File	Purpose
<code>.github/workflows/agent-cicd.yml</code>	GitHub Actions pipeline definition
<code>ai_agent.py</code>	LangChain agent using Gemini 2.5 Flash to decide GO/RETRY/STOP
<code>app.py</code>	Sample Python app with an intentional test failure
<code>requirements.txt</code>	Python dependencies (langchain-google-genai, pytest, etc.)
<code>deploy.sh</code>	Simple deploy script — copies <code>app.py</code> to <code>/opt/myapp/</code>
<code>test-results.log</code>	Generated during CI run — read by the AI agent
<code>decision.txt</code>	Written by the AI agent — read by the Decision Gate step

Troubleshooting

Pipeline not starting

Check that you pushed to main. Go to Actions tab and verify the workflow file exists at `.github/workflows/agent-cicd.yml`.

GEMINI_API_KEY not found

Confirm the secret is added inside the my-secrets environment (Settings → Environments → my-secrets → Environment secrets), not at repo or organisation level.

Agent Decision is always STOP

`app.py` has an intentional failing test (`assert divide(10,2)==4` is wrong). This is by design — the agent should return STOP. Fix the assertion to `assert divide(10,2)==5.0` to get a GO decision.

ModuleNotFoundError

The Install Dependencies step failed. Check the Actions log for the exact missing package and add it to `requirements.txt`.

Permission denied on `deploy.sh`

Add a `chmod` step before `deploy`: `run: chmod +x deploy.sh`